

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Information Technology Semester: III

Subject Code & Name: BTITC305 Data Structures and Applications

Max Marks: 60

Date:11/01/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What is the complexity of an algorithm? Define asymptotic notations for complexity of algorithms.	Understand Knowledge CO3	6
B) Describe the notion of the space-time tradeoff of algorithms. Calculate time complexity of the following function. int data(int list[], int n) { int i, sum= 0; for(i=0; i<n; i++) sum = sum + list[i] ; return sum ; }	Knowledge Apply CO3	6
C) Write an algorithm: i)To add two n x n matrices ii)To find the largest element in the list	Apply CO1	6
Q.2 Solve Any Two of the following.		12
A) Evaluate the following postfix expression using stack. 5, 6, 2, +, *, 12, 4, /, -	Evaluate CO2	6
B) What is a priority queue? Describe an array representation of a priority queue.	Knowledge Understand CO1	6
C) Design a procedure which deletes a node N with location LOC in a linked list.	Create CO1	6

Q.3 Solve Any Two of the following.

12

- A) Draw the tree structure for the following expression.

Apply

6

$$(x-4y)(6a-4b)^3$$

CO1

- B) Write a stack based algorithm for postorder traversal of a binary tree.

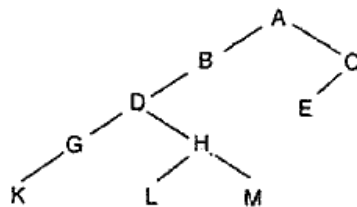
Create

Apply it on the following binary tree, show the steps.

Apply

CO3

6



- C) Define binary search tree.

Knowledge

6

Insert the following numbers in order into an empty binary search tree, show the stages.

Create

40, 60, 50, 33, 55, 11

CO1

Q.4 Solve Any Two of the following.

12

- A) Describe the following terms with reference to Graph.

Understand

6

i) Multigraph

CO1

ii) Strongly Connected Graph

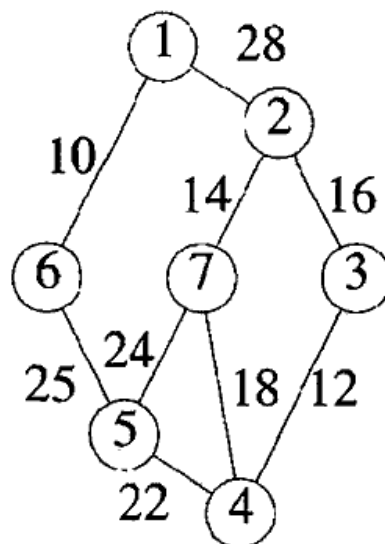
iii) Path Matrix <https://www.batuonline.com>

- B) Find the minimum spanning tree using Prim's algorithm for the following graph.

Apply

6

CO3



- C) Explain linked representation of a graph with a suitable example.

Understand

6

CO3

Q. 5 Solve Any Two of the following.

12

- | | | |
|--|---------------|----------|
| A) Write a C program to implement a Binary Search algorithm for searching a given item in the list. | Create
CO4 | 6 |
| B) Sort the following list using Selection Sort, show the passes.
77, 33, 44, 11, 88, 22, 66, 55 | Apply
CO3 | 6 |
| C) Write a C program to implement a Quick Sort algorithm for sorting a given list. | Create
CO5 | 6 |

***** End *****